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Supplementary Materials for

Artificial intelligence using a latent diffusion model enables the generation of diverse and potent antimicrobial peptides

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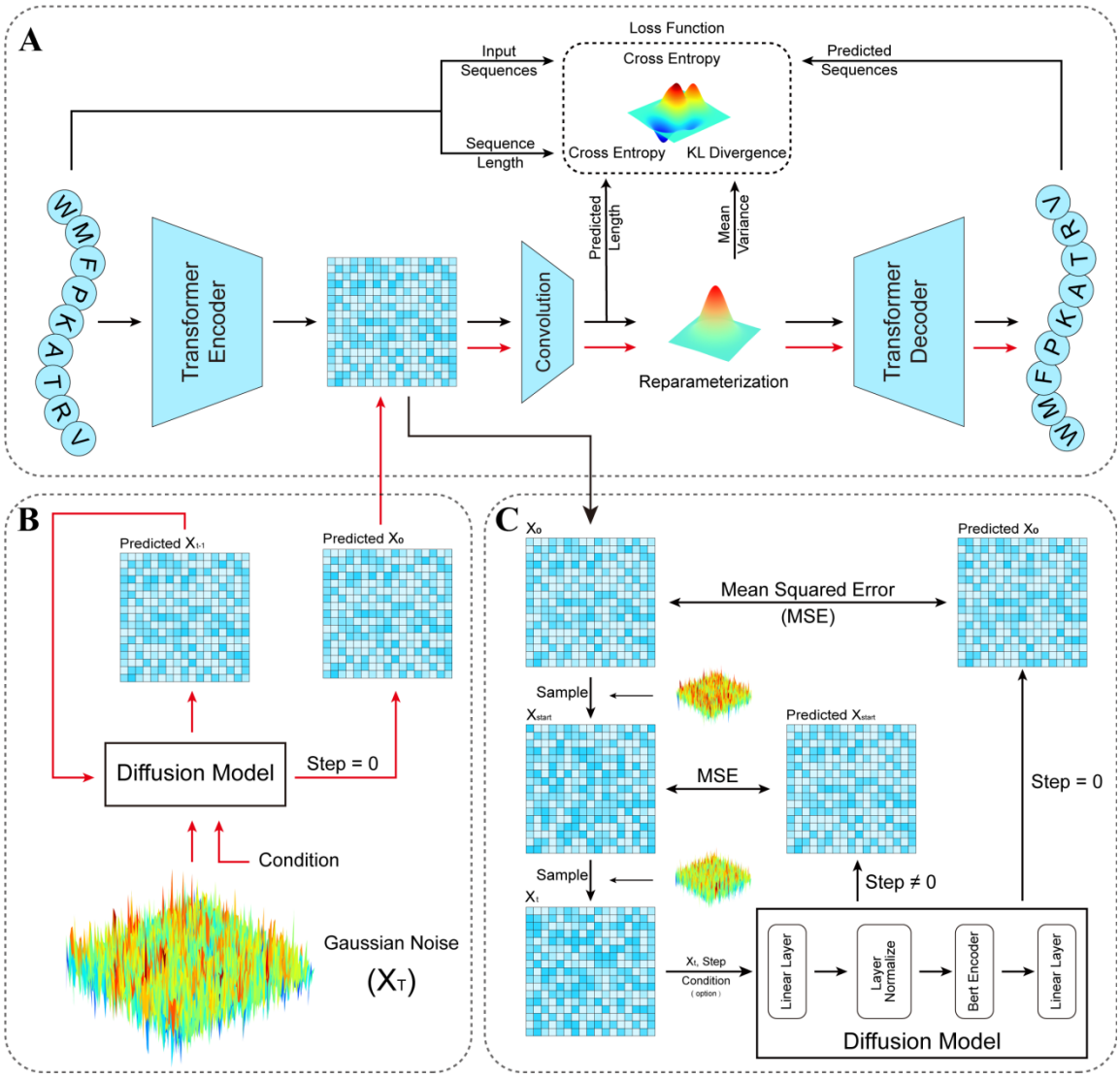


Fig. S1.

Overview of the latent diffusion architecture and data flow. (A) Data flow and optimization setup during VAE training. (B) Data flow during generation. (C) Data flow and optimization setup during diffusion model training. The black line represents the training step, and the red line represents the generation process.

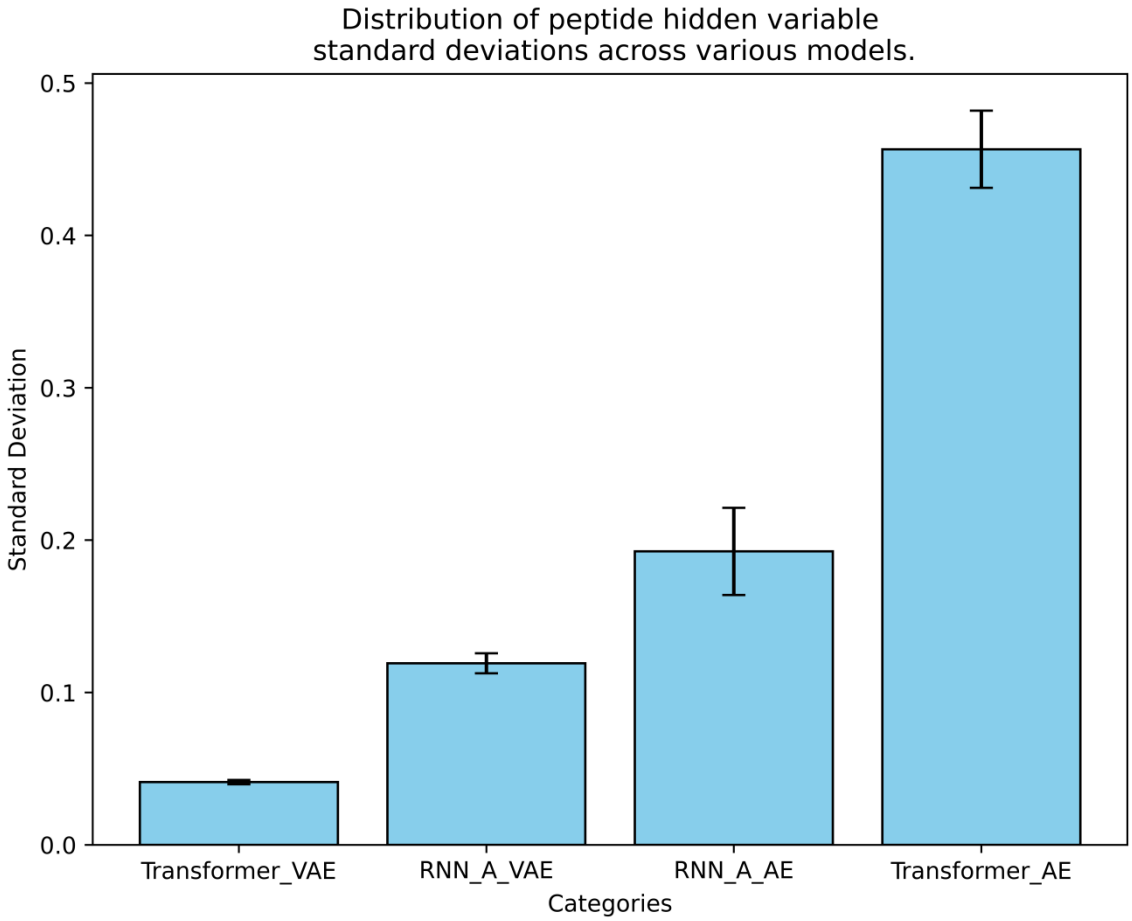


Fig. S2.

The distribution of peptide hidden variable standard deviations across various models. The terms 'Transformer_VAE' and 'Transformer_AE' refer to Variational Autoencoder (VAE) and Autoencoder (AE) models respectively, trained using a Transformer-based architecture. 'RNN_A_VAE' and 'RNN_A_AE' denote VAE and AE models trained on an RNN structure enhanced with an attention mechanism.

Our objective is to minimize the Euclidean distance (reconstruction error of diffusion model), as smaller Euclidean distances indicate greater proximity between the model's predictions and the actual values, thereby enhancing the accuracy of the model's forecasts. When $Loss_{MSE}$ value is similar, a larger scaling factor (α) results in a smaller Euclidean distance. As the standard deviations of hidden variables decrease, a larger scaling factor (α) is required. Therefore, we have opted for the Transformer_VAE model.

$$X_{scale}=X_{true}\times\alpha,\ X_{generate}=X_{predict}\times\alpha$$

$$\mathrm{Loss}_{\mathrm{MSE}}\left(X_{scale},X_{generate}\right)=\frac{1}{n}\sum_{i=1}^{n}\left(X_{scale,i}-X_{generate,i}\right)^2$$

$$\mathrm{d}\left(X_{true},X_{predict}\right)=\sqrt{\sum_{i=1}^{n}\left(X_{true,i}-X_{predict,i}\right)^2}=\frac{1}{\alpha}\sqrt{\sum_{i=1}^{n}\left(X_{scale,i}-X_{generate,i}\right)^2}=\frac{1}{\alpha}\sqrt{n\times Loss_{MSE}}$$

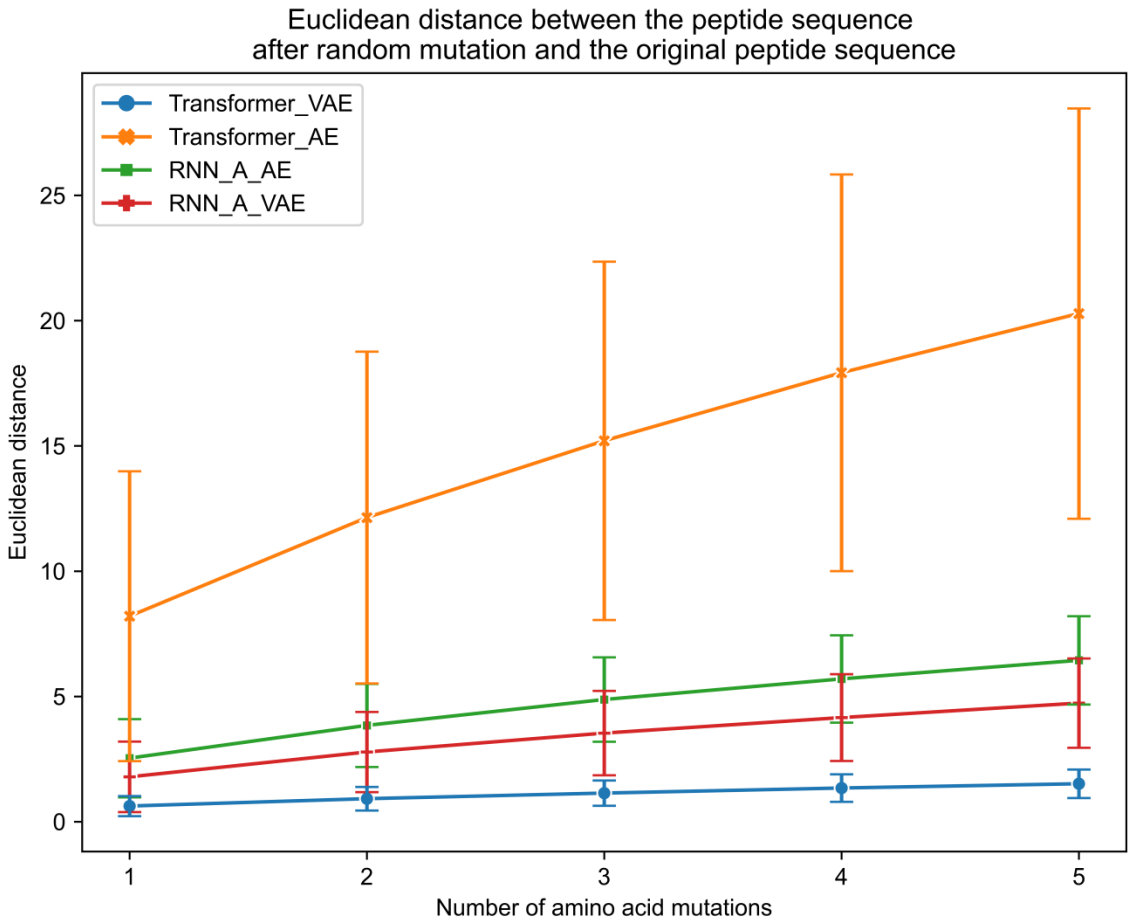


Fig. S3.

The Euclidean distance between the peptide sequence after random mutation and the original peptide sequence. The terms 'Transformer_VAE' and 'Transformer_AE' refer to Variational Autoencoder (VAE) and Autoencoder (AE) models respectively, trained using a Transformer-based architecture. 'RNN_A_VAE' and 'RNN_A_AE' denote VAE and AE models trained on an RNN structure enhanced with an attention mechanism.

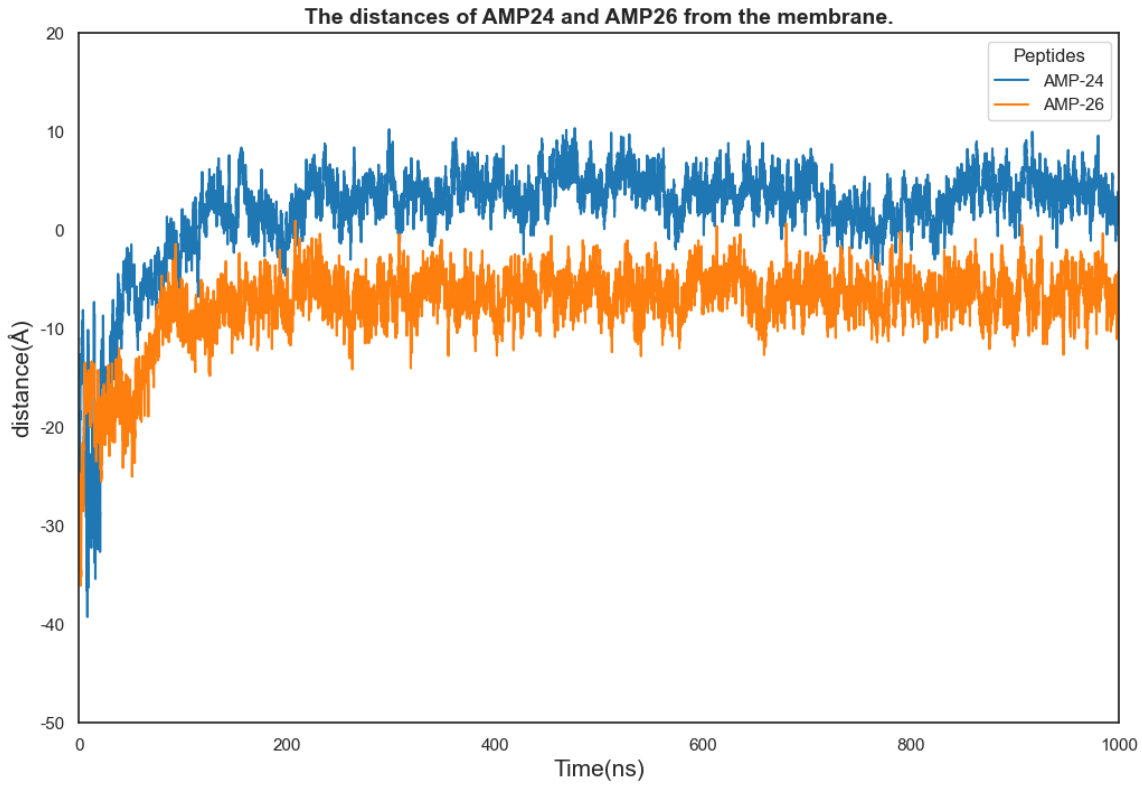


Fig. S4.
The distances of AMP24 and peptide 26 from the membrane.

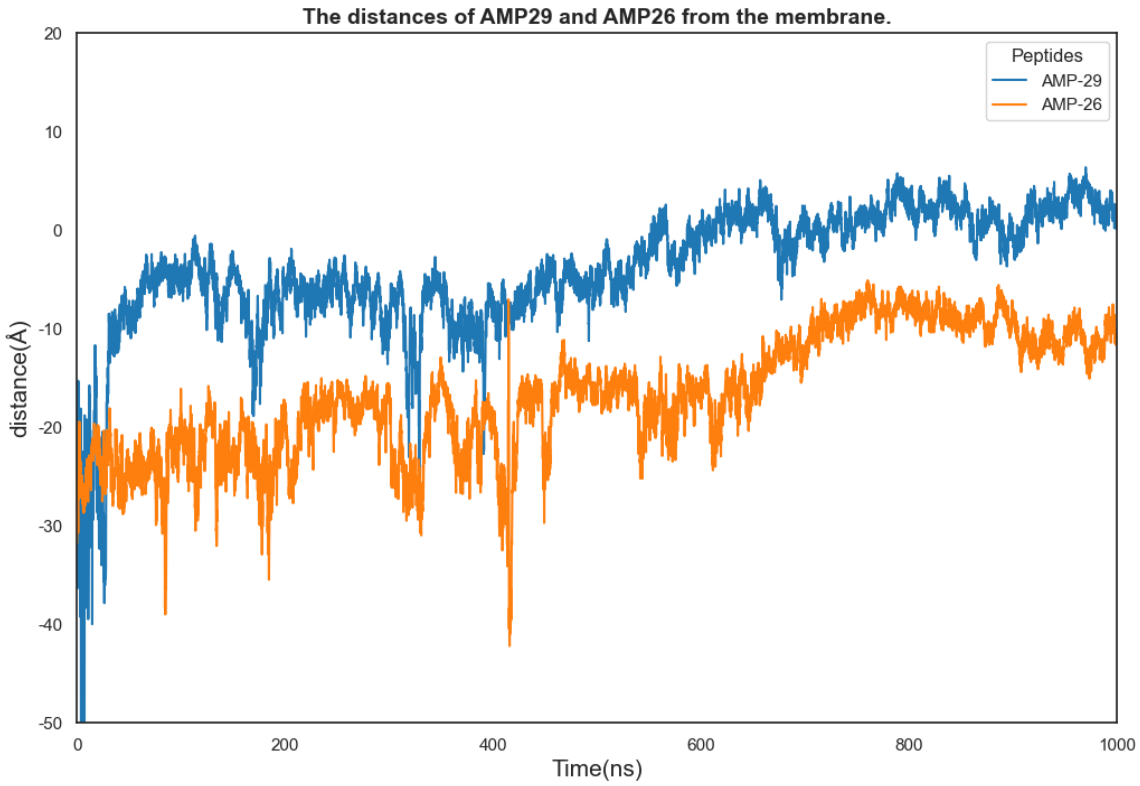


Fig. S5.
The distances of AMP29 and peptide 26 from the membrane.

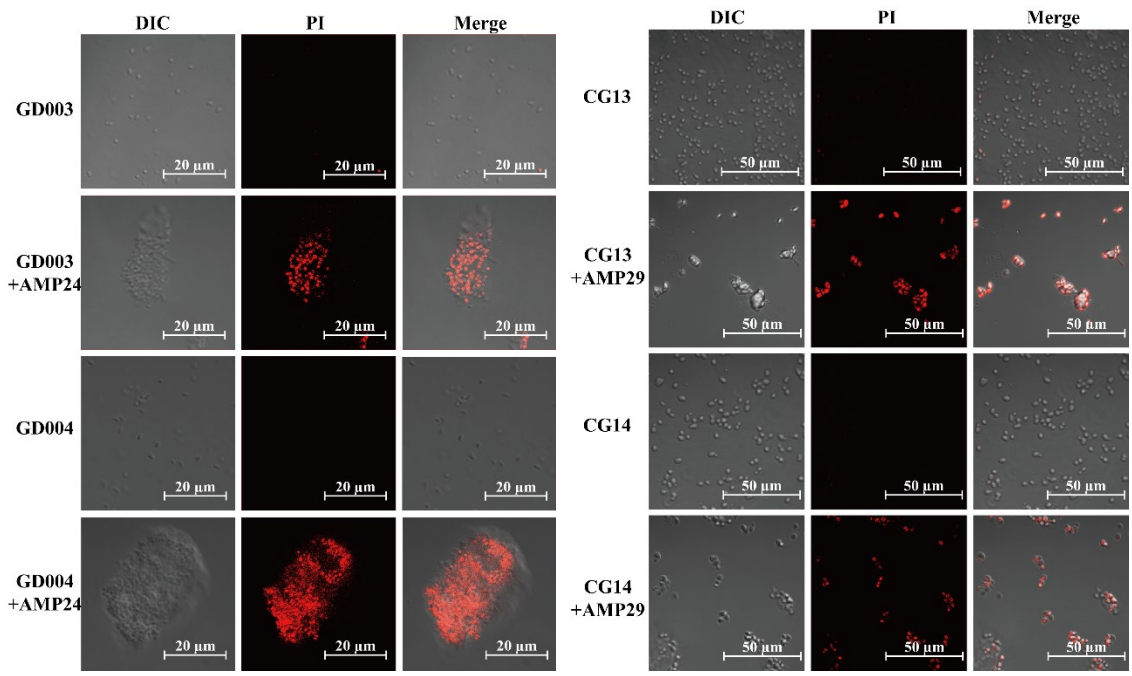


Fig. S6.

Analysis of cell membrane permeability. *A. baumannii* cultures of GD003 and GD004 were exposed to 50 μM of AMP-24 (a, scale bar, 20 μm) and *G. glabrata* cultures of CG13 and CG14 were subjected to 50 μM of AMP-29 (b, scale bar, 50 μm). After 3 hours treatment, cells were stained with 5 μM PI dye for 30 min and observed by confocal microscope.

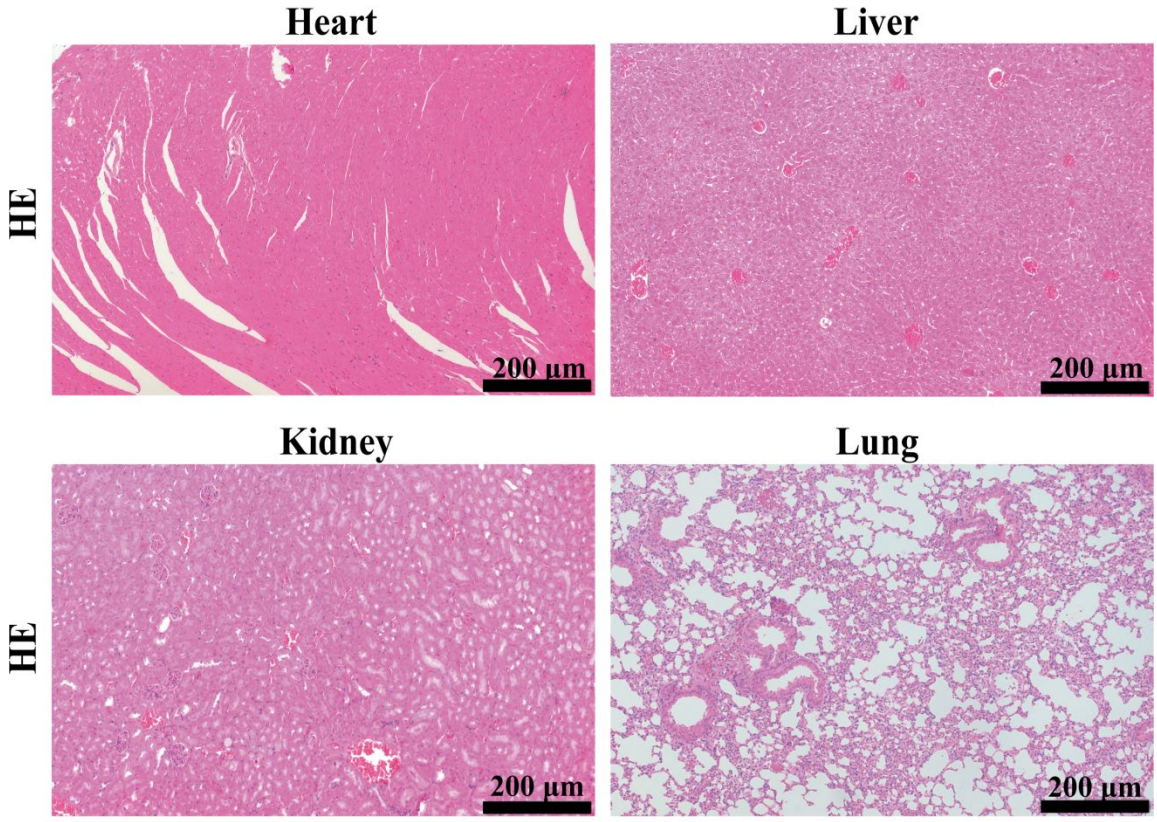


Fig. S7.
H&E staining images of heart, liver, kidney, and lung.

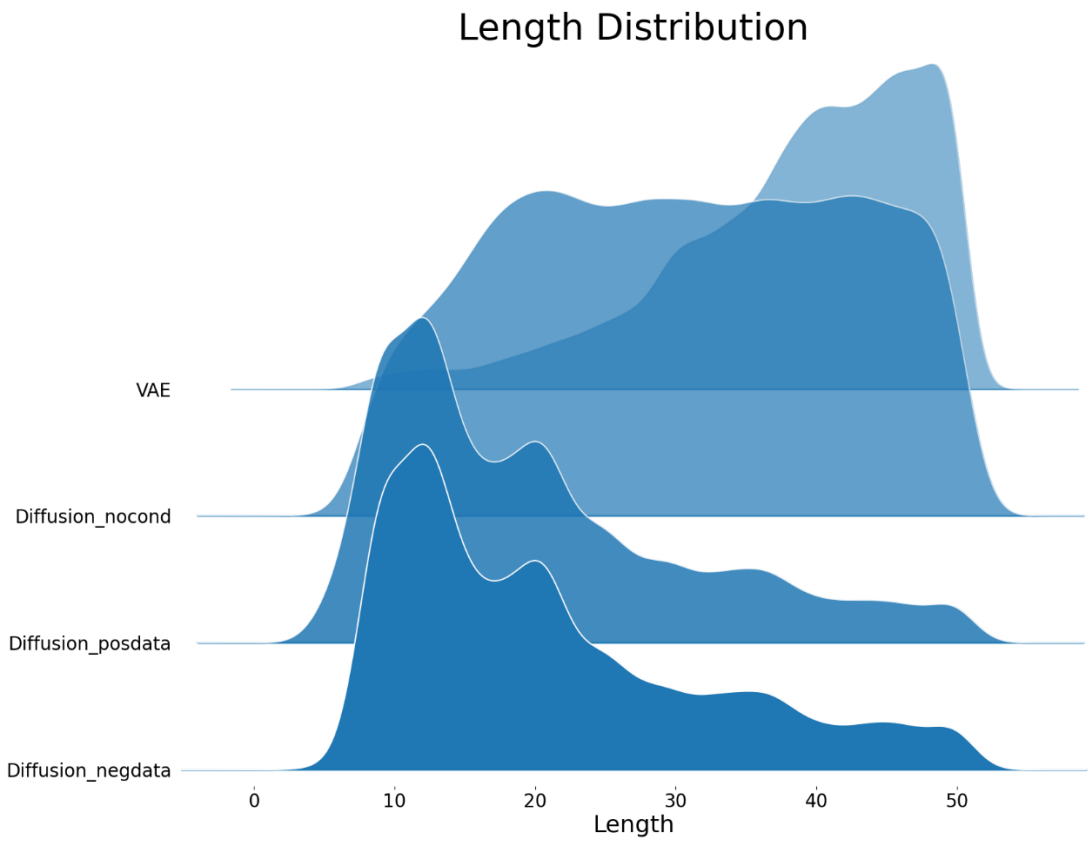


Fig. S8.

Amino acid length distributions of peptides used in different training sets. VAE means the amino acid length distributions of peptides used in VAE model. Diffusion_nocond means the amino acid length distributions of peptides used in pre-trained diffusion model. Diffusion_posdata and Diffusion_negdata represent the distribution of AMP and Non-AMP used to fine-tune the diffusion model, respectively.

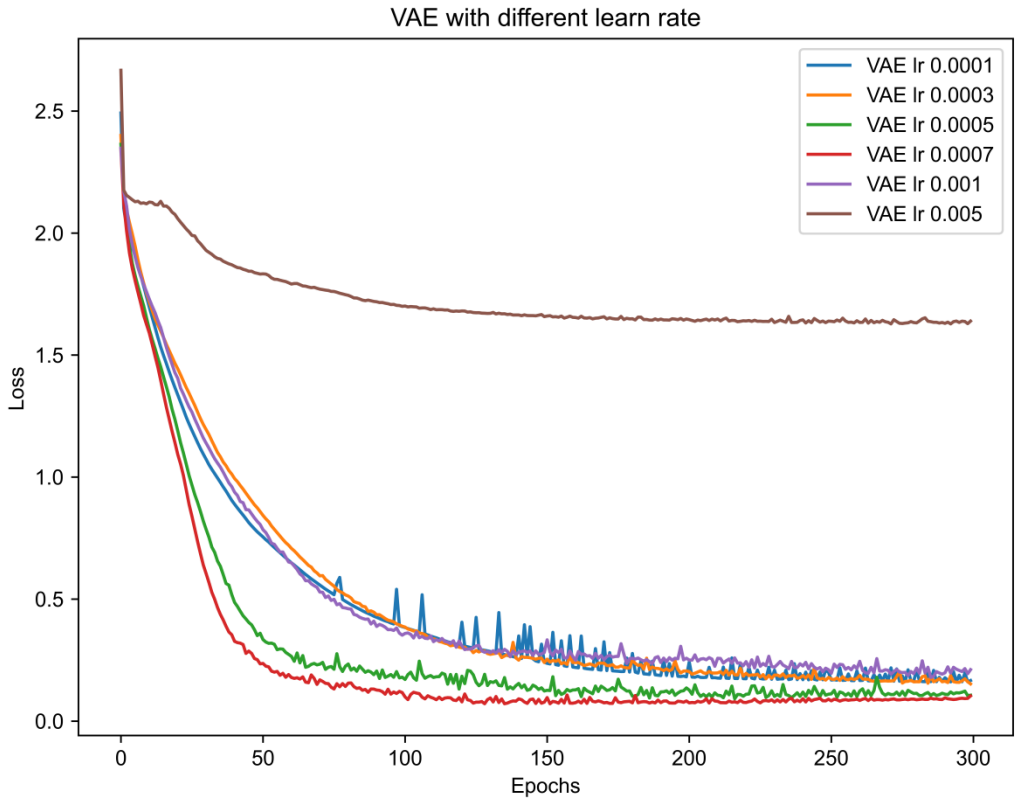


Fig. S9.
The loss during training process of VAE with different learn rate.

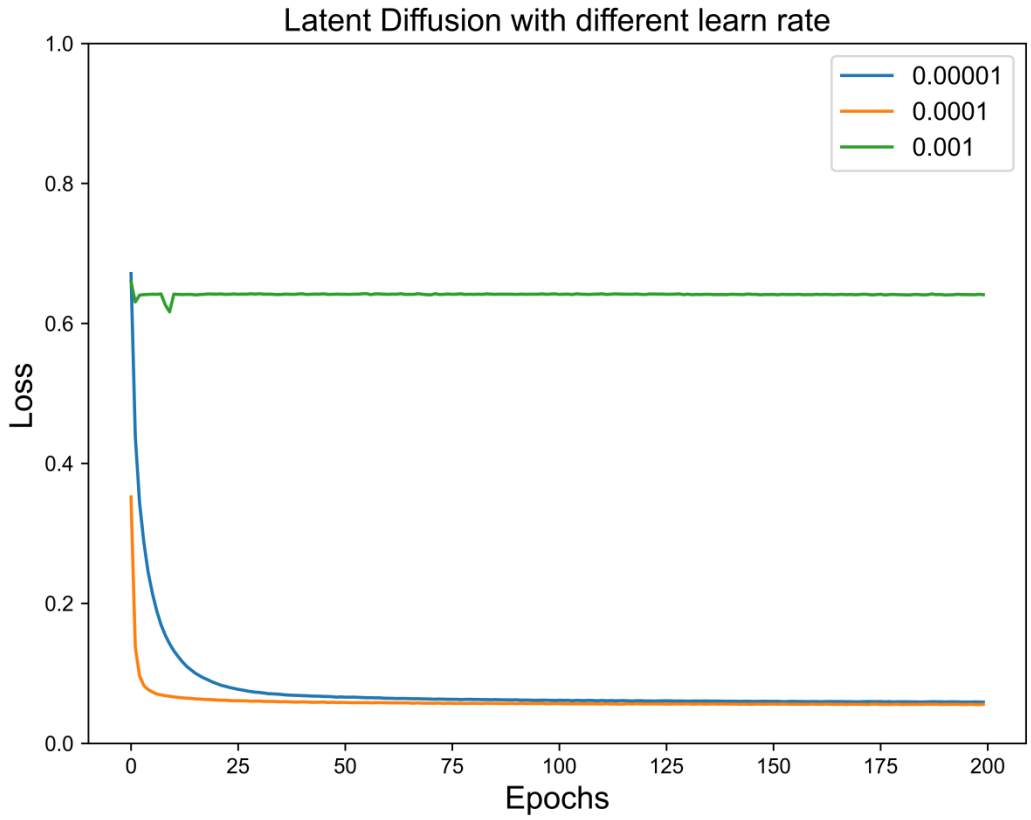


Fig. S10.
The loss during training process of Latent Diffusion with different learn rate.

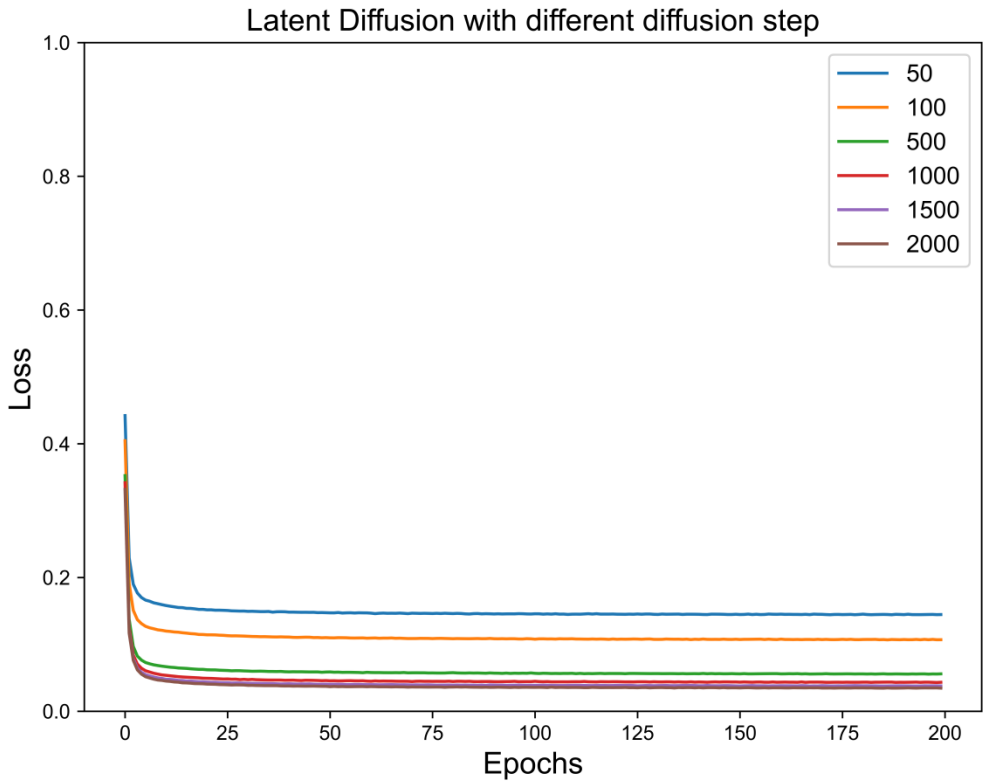


Fig. S11.
The loss during training process of Latent Diffusion with different diffusion step.

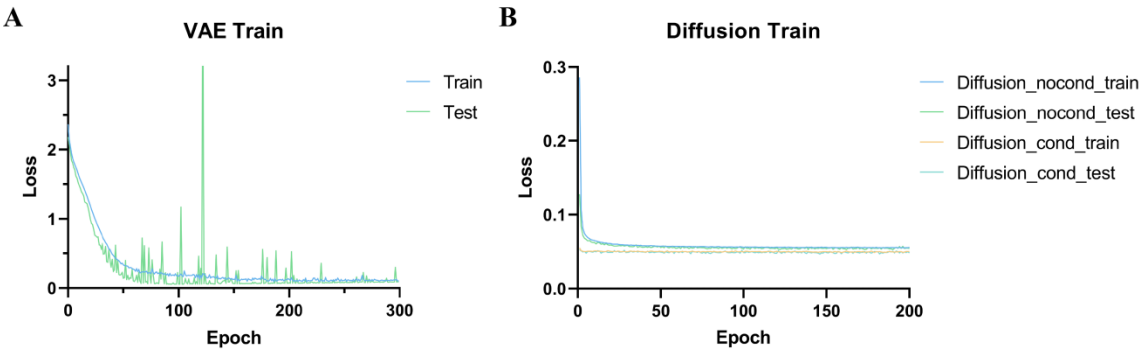


Fig. S12.

The loss during training and test process of VAE and diffusion models. (A) The loss of variational autoencoders (VAE). (B) The loss of diffusion model.

Table S1.

Top 10 ensembled models.

	Accuracy	AUC	AMP Precision	AMP Recall	AMP F1-score	Non-AMP Precision	Non-AMP Recall	Non-AMP F1-score
RCNN_10-fold RNN_attention_10-fold Transformer_10-fold	0.88935	0.882113	0.9926	0.7693	0.8668	0.8306	0.9949	0.9054
CNN_10-fold RNN_attention_10-fold Transformer_10-fold	0.88852	0.881306	0.9909	0.7689	0.8659	0.8302	0.9938	0.9046
RCNN_10-fold RNN_10-fold RNN_attention_10-fold'	0.865892	0.8571	0.9908	0.7201	0.834	0.8015	0.9941	0.8875
CNN_10-fold RCNN_10-fold RNN_attention_10-fold	0.897862	0.891315	0.9905	0.7893	0.8785	0.8428	0.9934	0.9119
CNN_10-fold RNN_10-fold RNN_attention_10-fold	0.865892	0.857154	0.9896	0.7209	0.8342	0.8019	0.9934	0.8874
RNN CNN_10-fold RNN_attention_10-fold	0.902429	0.896276	0.989	0.8004	0.8847	0.8496	0.9922	0.9154
RNN_attention CNN_10-fold RNN_attention_10-fold	0.900145	0.893836	0.989	0.7955	0.8817	0.8465	0.9922	0.9136
Transformer RCNN_10-fold RNN_attention_10-fold	0.894955	0.88829	0.9888	0.7844	0.8748	0.8396	0.9922	0.9095
RNN RCNN_10-fold RNN_attention_10-fold'	0.903882	0.897855	0.9885	0.8039	0.8867	0.8519	0.9918	0.9165

Table S2.

Comparison of the sequence similarity between the sequences generated by different models and their training sets.

	Number of sequences	Similarity	Identity
MLpep	28	0.7763 ± 0.0574	0.5887 ± 0.0499
Hydramp	26	0.7655 ± 0.1082	0.6298 ± 0.1126
CLASS	20	0.7499 ± 0.0814	0.6120 ± 0.1479
ML pepline	55	0.7662 ± 0.0571	0.5890 ± 0.0920
Our Model	40	0.5686 ± 0.0720	0.4195 ± 0.0609
Our Model (w/o cluster, class, biological filtering and cgmd)	5120	0.6032 ± 0.1194	0.4462 ± 0.1102
Our Model (w/o class, biological filtering and cgmd)	3917	0.5668 ± 0.1012	0.4130 ± 0.0911

w/o means without.

Table S3.

Intra-sequence similarity of AMPs used for activity validation.

	Similarity	Identity
MLpep	0.6845 ± 0.1121	0.5318 ± 0.1548
CLASS	0.6676 ± 0.1591	0.5308 ± 0.2300
ML pepline	0.9561 ± 0.0927	0.8192 ± 0.0971
Our Model	0.4162 ± 0.0626	0.2623 ± 0.0486
Our Model (w/o cluster, class, biological filtering and cgmd)	0.5686 ± 0.1272	0.4131 ± 0.5685
Our Model (w/o class, biological filtering and cgmd)	0.5349 ± 0.0999	0.3785 ± 0.0785

w/o means without.

Table S4.

Performance of random forest (Threshold=0.65).

	Precision	Recall	F1-score	Support
AMP	0.91	0.38	0.53	130
Non-AMP	0.66	0.97	0.79	164
Weight average	0.77	0.71	0.67	294

Table S5.
Validation of the activities of generated AMPs.

Compounds or peptides	MICs (μM)										
	C. albicans SC5314	*C. tropicalis* CT-Q-2	*C. glabrata* CG13	*C. parapsilosis* CP001	*C. neoformans* H99	*C. auris* CBS15108	*S. aureus* ATCC29213	*P. aeruginosa* GD001	*K. pneumoiae* GD002	*A. baumannii* GD003	*E. coli* GD004
Fluconazole	3.125	>400	50	3.125	25	>400	/	/	/	/	/
Oxacillin	/	/	/	/	/	/	25	/	/	/	/
Ampicillin	/	/	/	/	/	/	/	>400	>400	>400	>400
Piperacillin	/	/	/	/	/	/	/	>400	200	>400	>400
Amoxicillin	/	/	/	/	/	/	/	>400	>400	>400	>400
Cefotaxime sodium	/	/	/	/	/	/	/	>400	100	>400	>400
Ceftaroline fosamil	/	/	/	/	/	/	/	>400	400	>400	>400
AMP-1	>400	50	>400	>400	200	>400	400	>400	>400	>400	>400
AMP-2	>400	>400	>400	>400	50	>400	>400	**6.25**	50	**6.25**	25
AMP-3	100	>400	>400	>400	**6.25**	>400	>400	50	>400	>400	>400
AMP-4	>400	>400	>400	>400	>400	>400	>400	>400	>400	100	25
AMP-5	>400	>400	>400	>400	>400	>400	>400	100	>400	>400	>400
AMP-8	>400	>400	>400	>400	100	>400	>400	>400	>400	>400	>400
AMP-9	>400	50	>400	50	50	>400	>400	50	100	>400	25
AMP-11	>400	>400	>400	>400	**6.25**	>400	>400	**3.125**	**12.5**	**3.125**	**6.25**
AMP-12	>400	>400	>400	>400	**6.25**	>400	>400	>400	>400	>400	>400
AMP-13	>400	>400	>400	>400	>400	>400	>400	25	>400	>400	>400
AMP-14	>400	>400	>400	>400	100	>400	>400	50	>400	>400	>400
AMP-16	>400	200	>400	>400	>400	>400	>400	>400	>400	>400	>400
AMP-17	>400	>400	>400	>400	>400	>400	>400	**3.125**	**3.125**	**3.125**	**3.125**
AMP-20	>400	50	>400	>400	**6.25**	>400	>400	>400	>400	>400	>400
AMP-24	>400	100	>400	>400	200	>400	200	100	50	**6.25**	**6.25**
AMP-25	>400	100	>400	>400	>400	>400	>400	400	>400	**12.5**	50

AMP-29	>400	200	**6.25**	>400	>400	>400	>400	>400	>400	>400	>400
AMP-30	>400	200	>400	>400	400	>400	>400	>400	>400	>400	>400
AMP-31	>400	400	>400	>400	>400	>400	200	>400	>400	>400	>400
AMP-32	>400	50	>400	>400	>400	>400	>400	>400	>400	>400	>400
AMP-36	>400	>400	>400	>400	>400	>400	>400	>400	>400	>400	100
AMP-37	>400	50	>400	>400	>400	>400	>400	>400	>400	>400	>400
AMP-38	50	>400	>400	>400	>400	>400	>400	>400	>400	>400	>400
AMP-39	25	200	>400	50	>400	>400	>400	>400	>400	>400	>400
AMP-40	>400	>400	>400	>400	>400	>400	>400	25	>400	>400	50

a The bold of values means the potent activities of indicated AMPs against specific pathogens with its MIC values less-than or equal to 12.5 μM.
/ means non detected.

Table S6.
Antimicrobial peptides with excellent activity (μg/mL) corresponding to the results of Table 1 in the main text.

	C. glabrata CG13	*C. neoformans* H99	*P. aeruginosa* GD001	*K. pneumoniae* GD002	*A. baumannii* GD003	*E. coli* GD004
AMP-2	>400	178.66	22.33	178.66	22.33	89.33
AMP-3	>400	19.59	156.74	>400	>400	>400
AMP-11	>400	21.16	10.58	42.31	10.58	21.16
AMP-12	>400	29.48	>400	>400	>400	>400
AMP-17	>400	>400	13.37	13.37	13.37	13.37
AMP-20	>400	20.27	>400	>400	>400	>400
AMP-24	>400	527.84	263.92	131.96	16.50	16.50
AMP-25	>400	>400	871.84	>400	27.25	108.98
AMP-29	10.37	>400	>400	>400	>400	>400

Table S7.

Evaluation of AMP-29 against 9 additional *C. glabrata* strains.

	MICs (μM)								
Compounds or peptides	*C. glabrata*								
	CG2	CG3	CG4	CG8	CG11	CG12	CG14	CG15	CG17
Fluconazole	200	25	3.125	3.125	3.125	3.125	100	6.25	6.25
AMP-29	6.25	6.25	6.25	12.5	12.5	6.25	3.125	6.25	6.25

Table S8.

The antifungal activities of AI-designed antifungal peptides compared to (pre-)clinical phase peptides.

	MICs (μM)		
	C. glabrata	*C. albicans*	*C. tropicalis*
LL-37	>200	>200	>200
AMP-29	3.125~12.5	>400	200

Table S9.

Further evaluations of peptides 3, 11, 12 and 20 against another five *C. neoformans* strains.

	MICs (μM)				
Compounds or peptides	*C. neoformans*				
	108	117	129	134	138
Fluconazole	25	50	50	25	25
AMP-3	6.25	6.25	6.25	6.25	6.25
AMP-11	12.5	12.5	12.5	12.5	6.25
AMP-12	6.25	6.25	6.25	6.25	6.25
AMP-20	6.25	6.25	6.25	6.25	6.25

Table S10.

The antibacterial activities of AI-designed antibacterial peptides compared to (pre-)clinical phase peptides.

	MICs (μM)		
	P. aeruginosa	*A. baumannii*	*E. coli*
Iseganan	3.125	3.125	6.25
LL-37	6.25	3.125	6.25
AMP-2	6.25	1.562~3.125	25
AMP-11	3.125	1.562~3.125	3.125~6.25
AMP-17	3.125	3.125	1.562~3.125
AMP-24	100	6.25	6.25

Table S11.

The antibacterial activities of five peptides (AMP-2, AMP-11, AMP-17, AMP-24 and AMP-25) against *A. baumannii* and *E. coli* strains.

Compounds or peptides	MICs (μM)								
	A. baumannii				*E. coli*				
	GD 4948	GD 4906	GD 4955	GD 144013	GD 7188	GD 7135	GD 7465	GD 6738	GD 7354
Ampicillin	200	100	50	100	12.5	25	50	>400	>400
Piperacillin	200	100	100	100	50	50	100	100	>400
Amoxicillin	50	25	3.125	25	6.25	12.5	100	>400	>400
Cefotaxime sodium	6.25	3.125	12.5	6.25	1.562	<0.75	6.25	400	>400
Ceftaroline fosamil	12.5	12.5	12.5	12.5	<0.75	<0.75	12.5	>400	>400
AMP-2	3.125	1.562	1.562	3.125	/	/	/	/	/
AMP-11	1.562	3.125	3.125	3.125	3.125	3.125	6.25	3.125	3.125
AMP-17	3.125	3.125	3.125	3.125	1.562	3.125	3.125	1.562	1.562
AMP-24	12.5	12.5	12.5	12.5	25	25	12.5	12.5	12.5
AMP-25	25	25	25	12.5	/	/	/	/	/

/ means non detected.

Table S12.

Cytotoxicity of AMPs against mammalian cells.

	IC_{50} (μM)				
Cell strains	Antibacterial peptides				
	AMP-2	AMP-11	AMP-17	AMP-24	AMP-25
HaCaT	47.17	23.89	29.39	>200	31.39
HUVEC	22.64	13.10	36.23	>200	39.48
Cell strains	Antifungal peptides				
	AMP-3	AMP-11	AMP-12	AMP-20	AMP-29
HaCaT	7.16	23.89	4.82	7.09	>200
HUVEC	8.33	13.10	16.87	5.62	>200

Table S13.

Performance of various autoencoders.

Architecture	BLEU↑	Reconstruction Error↓	Token Accuracy↑
Our Model	**0.998**	**0.008**	**0.9993**
WAE[1]	0.892	0.216	
HydrAMP(temp=0)[2]	0.868	0.534	0.9271

[1, 2] The results of WAE model were reported by the authors in reference 1. The results of the HydrAMP model were calculated using the code released by the authors in reference 2.

Table S14.

Amino acids distribution

	AMPs in training sets	Generated AMPs	Non-AMPs in training sets
G	0.075995	0.077132	0.060967
A	0.069131	0.061255	0.069155
L	0.094128	0.102080	0.098940
M	0.011416	0.011206	0.053204
F	0.010954	0.042756	0.044820
W	0.039568	0.032117	0.014334
K	0.120705	0.104847	0.059228
Q	0.026624	0.033328	0.036530
E	0.027482	0.026005	0.052176
S	0.044819	0.046677	0.074694
P	0.036306	0.037134	0.045086
V	0.061631	0.057017	0.061536
I	0.064254	0.065676	0.059725
C	0.052230	0.048695	0.021792
Y	0.023431	0.026361	0.028975
H	0.021635	0.027187	0.023832
R	0.100549	0.104338	0.061796
N	0.034680	0.039190	0.039875
D	0.022000	0.023545	0.040793
T	0.032462	0.033453	0.052542

Table S15.

The number of amino acids and weight of AMPs that are used for the embedding layer.

Name	Number	Weight
<start>	0	None
E	1	0.83202248
I	2	0.81670196
F	3	0.85313521
A	4	0.79636116
K	5	0.81604869
N	6	0.86721444
G	7	0.81542819
M	8	0.8660451
T	9	0.83560374
P	10	0.85635282
L	11	0.76951772
R	12	0.81529956
D	13	0.85887284
S	14	0.80742307
V	15	0.81418058
Y	16	0.90343196
Q	17	0.88166984
C	18	0.95441499
W	19	1.0
H	20	0.93364147
<pad>	21	0.1
<end>	22	0.91921199

Table S16.

Hyperparameters for different classifiers.

Name	Batch Size	Learning rate	Epochs	Evaluation
CNN	512	0.00001	150	accuracy
RNN	512	0.001	150	accuracy
RNN attention	512	0.0001	150	accuracy
RCNN	512	0.00001	150	accuracy
Transformer	512	0.000001	150	accuracy

Table S17.

Hyperparameters for random forest

Hyperparameters	Random Forest
N_estimators	50
Max_depth	9
Max_features	5
Min_samples_leaf	7
Bootstrap	True
Criterion	entropy

Table S18.
Scores for the top-40 peptides

ID	Sequences	Score
1	RFRDVVRIISTGVHSVAR	0.9573
2	GLGSLVHGVKGRIFNHKLGMASVIKGVLQKLWE	0.9539
3	IGGKNGGWWGKVRWGLLSLIKGIAGQRLL	0.9532
4	LVSGLNSKRFFGLKGGNLFKKALKNVSVFGKAGVTSALR	0.9467
5	YQRVLRRLPRQFFPRPQPRTPWL	0.9467
6	PYWQTGSGVRLVP	0.9457
7	FLGPLVKNLSIELGKSIKRIVSKILGEVLQAYF	0.9429
8	LTWPKFSQRIRPAGPIGKIAHHLTVHIIPQLI	0.9397
9	GIGTVLRGWLKGVMTPVIGLKHSL	0.9391
10	GDWNWINGLATVVGGVIGKIVTITISIKY	0.9385
11	GIWDMIRKVRTVIRKAKQIKTFINGKGVQ	0.9371
12	GWRTVLKFGWVGKLYWGVAKAANKMARYLTKAVYGRVRKDA	0.9368
13	GIIEKLLKGIGKVVSGVLKATAGAQSGGSQIKASQKIAS	0.9355
14	GLLSRLLGAKARAGKQALRNTAGQIRRGLLGARAAP	0.9318
15	GIGSVAHSAVKAKAQAAEQLAGAGQAGHGINHWVG	0.9317
16	SVRTVGARAFGAVSRVW	0.9297
17	AKLGKTIKKLLKSKKFKIRGELLQGLRKASVQLGTLGIL	0.9281
18	GLPPWVNGPTWAR	0.9268
19	GRFRNVGGEHHAGLIGWIKAARIRVTAL	0.9252
20	GFGRFWRSVISGLAKNFLGKLREISQLY	0.9211
21	VQQLLTGLQRALN	0.9196
22	YLPQGVQQYAFRW	0.9107
23	FITNYGNFGIRIRGILRK	0.9093
24	WLGSKIKAAKMKVGMWGAVGLRHN	0.9061
25	GIWKPIFTYLATPIHFKF	0.8975
26	VTFVRLAGVASAVKTKAG	0.8954
27	AMPGKVSIVTAGQIVGAKTGKAL	0.8908
28	GFLPAFGLAPGNDAPYIA	0.8844
29	WFHFFQRGSYNGI	0.8819
30	ILPTKWWSWPWIRISW	0.8792
31	ARGSILRQALELHGKLLN	0.8727
32	FIHMFHTLAKFSF	0.8706
33	WLPLKYPQLMSAP	0.8673
34	GFGRGLIRVSGKILGEAGYGGAG	0.8658
35	IVPFHPDWLPGGW	0.8649
36	ARFGLQLGTAGGLLTIGASMIQKL	0.8431
37	ALSQIWKIHSKVP	0.8353
38	GDLKNIYSFLKQALPGLL	0.8292
39	LKPIRIRIWQVIKPILRI	0.8122
40	LKFLKALKDELKSAGSAIGKVLHK	0.8053

Table S19.
Physicochemical properties of the top-40 peptides

ID	Sequences	Length	Maximal Hydrophobic Moment	Isoelectric Point	Aliphatic Index	Net Charge	Hydrophobicity Index
1	RFRDVVRIISTGVHSVAR	18	0.8265	12.5001	113.3333	3.0893	0.1667
2	GLGSLVHGVKGRIFNHKLGMASVIKGVLQKLWE	33	0.5367	11.2482	120.9091	4.1804	0.2848
3	IGGKNGGWWGKVRWGLLSLIKGIAGQRLL	29	0.4026	12.5461	121.0345	4.9971	0.1414
4	LVSGLNSKRFFGLKGGNLFKKALKNVSVFGKAGV TSALR	39	0.7006	12.5846	97.4359	7.9962	0.1795
5	YQRVLRRLPRQFFPRPQPRTPWL	23	1.0613	12.8017	63.4783	5.9971	-1.1826
6	PYWQTGSGVRLVP	13	0.2851	9.3488	74.6154	0.9971	-0.2692
7	FLGPLVKNLSIELGKSIKRIVSKILGEVLQAYF	33	0.8551	10.4812	147.5758	2.9995	0.6121
8	LTWPKFSQRIRPAGPIGKIAHHLTVHIIPQLI	32	0.5808	12.5317	125.0000	4.2701	0.1656
9	GIGTVLRGWLKGVMTPVIGLKHSL	24	0.6441	11.8231	133.7500	3.0883	0.6875
10	GDWNWINGLATVVGGVIGKIVTITISIKY	29	0.4336	9.2560	137.5862	0.9970	0.7966
11	GIWDMIRKVRTVIRKAKQIKTFINGKGVQ	29	0.8957	12.2964	100.6897	6.9969	-0.3069
12	GWRTVLKFGWVGKLYWGVAKAANKMARYLTKA VYGRVRKDA	41	0.7534	11.4175	78.5366	8.9941	-0.2561
13	GIIEKLLKGIGKVVSGVLKATAGAQSGGSQIKASQ KIAS	39	0.7937	11.1021	115.1282	4.9980	0.2718
14	GLLSRLLGAKARAGKQALRNTAGQIRRGLLGARA AP	36	0.7334	13.2031	108.8889	7.9974	-0.1472
15	GIGSVAHSAVKAKAQAAEQLAGAGQAGHGINHW VG	35	0.4779	9.5435	84.0000	1.2719	0.0400
16	SVRTVGARAFGAVSRVW	17	0.6616	12.8011	85.8824	2.9980	0.4412
17	AKLGKTIKKLLKSKKFKIRGELLQGLRKASVQLGT LGIL	39	0.8510	12.0273	132.5641	9.9971	-0.0333
18	GLPPWVNGPTWAR	13	0.2593	10.5500	60.0000	0.9980	-0.4846
19	GRFRNVGGEHHAGLIGWIKAARIRVTAL	28	0.3417	12.5084	104.6429	4.1813	0.0214
20	GFGRFWRSVISGLAKNFLGKLREISQLY	28	0.7497	11.5768	97.5000	3.9983	0.0214
21	VQQLLTGLQRALN	13	0.6650	10.5500	150.0000	0.9980	0.1231
22	YLPQGVQQYAFRW	13	0.4374	9.1716	60.0000	0.9963	-0.6077
23	FITNYGNFGIRIRGILRK	18	0.5673	12.2271	108.3333	3.9968	-0.0111
24	WLGSKIKAAKMKVGMWGAVGLRHN	24	0.3594	11.9974	85.4167	5.0877	-0.0542
25	GIWKPIFTYLATPIHFKF	18	0.4306	10.2477	92.2222	2.0875	0.5167
26	VTFVRLAGVASAVKTKAG	18	0.3325	11.8231	108.3333	2.9974	0.8500
27	AMPGKVSIVTAGQIVGAKTGKAL	23	0.2431	11.1031	106.0870	2.9971	0.6043
28	GFLPAFGLAPGNDAPYIA	18	0.1705	3.7500	87.2222	-1.0024	0.5889
29	WFHFFQRGSYNGI	13	0.4920	9.3512	30.0000	1.0880	-0.4308
30	ILPTKWWSWPWIRISW	16	0.3376	11.6518	97.5000	1.9977	-0.0688
31	ARGSILRQALELHGKLLN	18	0.5648	11.4797	141.1111	2.0904	-0.0611
32	FIHMFHTLAKFSF	13	0.5207	9.7040	67.6923	1.1795	0.8769
33	WLPLKYPQLMSAP	13	0.1897	9.2983	97.6923	0.9968	-0.0077
34	GFGRGLIRVSGKILGEAGYGGAG	23	0.7286	10.4465	89.1304	1.9986	0.2217
35	IVPFHPDWLPGGW	13	0.2527	5.2923	82.3077	-0.9107	0.0923
36	ARFGLQLGTAGGLLTIGASMIQKL	24	0.4446	11.6518	126.2500	1.9977	0.7708
37	ALSQIWKIHSKVP	13	0.5679	10.8054	120.0000	2.0883	0.0154
38	GDLKNIYSFLKQALPGLL	18	0.6525	9.2560	135.5556	0.9970	0.2944
39	LKPIRIRIWQVIKPILRI	18	0.8785	12.8136	189.4444	4.9974	0.5500
40	LKFLKALKDELKSAGSAIGKVLHK	24	0.6994	10.8018	122.0833	4.0893	-0.0042